

Chatbot Artificial Intelligence for Genetic Cancer Risk Assessment and Counseling: A Systematic Review and Meta-Analysis

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ABSTRACT

PURPOSE Most individuals with a hereditary cancer syndrome are unaware of their genetic status to underutilization of hereditary cancer risk assessment. Chatbots, or programs that use artificial intelligence to simulate conversation, have emerged as a promising tool in health care and, more recently, as a potential tool for genetic cancer risk assessment and counseling. Here, we evaluated the existing literature on the use of chatbots in genetic cancer risk assessment and counseling.

METHODS A systematic review was conducted using key electronic databases to identify studies which use chatbots for genetic cancer risk assessment and counseling. Eligible studies were further subjected to meta-analysis.

RESULTS Seven studies met inclusion criteria, evaluating five distinct chatbots. Three studies evaluated a chatbot that could perform genetic cancer risk assessment, one study evaluated a chatbot that offered patient counseling, and three studies included both functions. The pooled estimated completion rate for the genetic cancer risk assessment was 36.7% (95% CI, 14.8 to 65.9). Two studies included comprehensive patient characteristics, and none involved a comparison group. Chatbots varied as to the involvement of a health care provider in the process of risk assessment and counseling.

CONCLUSION Chatbots have been used to streamline genetic cancer risk assessment and counseling and hold promise for reducing barriers to genetic services. Data regarding user and nonuser characteristics are lacking, as are data regarding comparative effectiveness to usual care. Future research may consider the impact of chatbots on equitable access to genetic services.

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INTRODUCTION

An estimated 1.1% of individuals in the United States carry a cancer-predisposing germline pathogenic variant.¹ Such pathogenic variants can markedly increase lifetime cancer risk, and identification of individuals harboring cancer-predisposing pathogenic variants can allow for the implementation of personalized risk reduction strategies, targeted surveillance, and tailored treatment options that reduce cancer incidence, morbidity, and mortality.²⁻⁵ Despite the endorsement of hereditary cancer screening at a primary care level by multiple national organizations, only an estimated 5% of individuals harboring cancer-predisposing pathogenic variants are aware of their genetic status.^{6,7} This is largely driven by the burden associated with genetic cancer risk assessment at the population level and challenges associated

with the facilitation of genetic testing among eligible individuals.⁸⁻¹¹ Efforts to reduce barriers to these processes may improve access to genetic services, allowing for patients to receive appropriate risk-reducing care.

Chatbots use artificial intelligence and natural language processing to mimic regular human interaction, either through verbal or written language.¹² Recently, such technology has emerged as a promising tool for streamlining medical processes. Chatbots have been used for purposes such as symptom monitoring, information collection, risk assessment, and education.¹³⁻¹⁷ Utilization of health information technology has demonstrated efficacy in collection of family health history—a cornerstone for genetic cancer risk assessment.¹⁸⁻²⁰ This is particularly advantageous for automatically identifying high-risk family history features that

meet established criteria for cancer genetic testing, such as those outlined by the National Comprehensive Cancer Network.^{21,22} Although these criteria are comprehensive and continually updated, it can be challenging for clinicians to stay abreast of the most current guidelines, making it difficult for them to accurately conduct genetic cancer risk assessments.^{23,24} In a similar vein, artificial intelligence–based tools have also been used for counseling and education across multiple areas of medicine.^{25,26} With innate analytic capability and the ability to interact with individuals in a multitude of settings, chatbots may complement the existing framework for risk assessment and education for hereditary cancer syndromes.²⁷

We sought to perform a systematic review and meta-analysis of the literature on use of chatbot artificial intelligence for genetic cancer risk assessment and counseling.

METHODS

Overview

This study was preregistered with PROSPERO (registration ID CRD42022351367). Exemption status was received from the institutional review board committee. The comprehensive literature search was conducted on September 12, 2022. A systematic search of online databases (Embase, MEDLINE, the Cochrane Central Register of Controlled Trials, Web of Science, Engineering Village, IEEE Xplore, and the ACM Digital Library) was performed (Data Supplement, Material S1, also available at OSF²⁸). All original articles meeting search criteria were included. No other language, publication date, publication status, or article type restrictions were used. In addition to database searches, Cochrane Handbook recommendations were followed to locate unpublished studies.²⁹ ClinicalTrials.gov and WHO International Clinical Trials Registry Platform were searched to locate unpublished trials. Reference checking of included studies was performed using Scopus. ProQuest Dissertations and Theses was searched for dissertations. Websites of selected government agencies and nongovernmental organizations were searched for relevant reports. Study authors were contacted as needed for clarification regarding eligibility criteria. All records were imported into Covidence systematic review management system (Covidence³⁰).

Inclusion Criteria

Studies were eligible if they met the following criteria designed around the PICO format (Data Supplement, Material S2): Population: individuals undergoing genetic cancer risk assessment via personal or family history screening for the purposes of identifying individuals at risk for hereditary cancer syndromes and/or individuals receiving pretest genetic counseling for hereditary cancer syndrome testing. Intervention: the use of a chatbot to perform genetic cancer risk assessment to identify individuals at risk for hereditary cancer syndromes and/or the use of a chatbot to perform pretest genetic counseling for hereditary cancer syndrome testing. A

tool was considered a chatbot if described as such within the study. Additionally, given the heterogeneous terminology, a tool was considered a chatbot if the tool adhered to chatbot criteria set forth by McTear et al,³¹ namely that a chatbot received language input by users, independently interpreted language, and produced language output.¹⁴ Comparison: No comparison group was required. Outcome: Outcomes included identification of chatbots that screened for hereditary cancer or performed pretest genetic counseling, as well as the proportion identified as high risk for hereditary cancer, the proportion receiving genetic testing, and the proportion found to be positive for a hereditary cancer syndrome. Additional outcomes included chatbot completion, study population characteristics, patient knowledge after use of chatbot for counseling, time required to complete chatbot, satisfaction with chatbot, and chatbot language availability. Duplicate publications of any chatbot reporting on the same patient cohort as another were excluded from analysis, with preference for inclusion given to peer-reviewed journal articles.

Data Extraction

Two investigators independently reviewed titles and abstracts to determine whether the studies warranted further screening for eligibility. Full-text versions, if available, were then independently reviewed by two investigators for inclusion. Discrepancies were resolved by discussion among investigators.

Statistical Analysis

Meta-analyses for the proportion of individuals completing the genetic cancer risk assessment portion of the chatbots, individuals identified to meet criteria for genetic testing by the chatbot, individuals who were eligible and ordered genetic testing, and individuals with cancer-predisposing pathogenic variants were conducted with the use of R software (version 3.6.1 [July 5, 2019], R Foundation for Statistical Computing, Vienna, Austria). Statistical heterogeneity was tested through the chi-square test (ie, Cochrane Q test), and a P value $\leq .20$ was used to indicate the presence of heterogeneity. Statistical heterogeneity was also assessed by the inconsistency statistic (I²). However, regardless of the heterogeneity test P value or I² statistic percentage, a random-effects analysis was used to calculate the pooled proportions. The random-effects analysis allows for more variability in the individual study proportion estimates when generating the pooled proportion and is more conservative. The pooled proportion was calculated using the logit transformation, and the 95% CI was calculated using the Clopper-Pearson interval, also called the exact binomial interval. To estimate the between-study variance, the maximum likelihood estimator was used.

Risk of Bias Assessment

The Joanna Briggs Institute Critical Appraisal Checklist for cross-sectional studies was used to assess the use of a

chatbot for the purpose of genetic cancer risk assessment or counseling.³² Risk of bias assessments were conducted by two independent investigators, and discrepancies were resolved by discussion among investigators.

RESULTS

Seven-hundred and forty records were identified which yielded 549 potentially relevant records after removal of duplicates (Fig 1). After title and abstract review, 36 items were eligible for full-text review, if available. Seven studies, evaluating five different chatbots, met inclusion criteria. Three studies³³⁻³⁵ included chatbots that were described to perform personal and family history-based genetic cancer risk assessments, one study³⁶ included a chatbot that was described to perform pretest hereditary cancer genetic testing counseling, and three studies³⁷⁻³⁹ included chatbots that were described to perform both risk assessment and counseling (Table 1). Five studies³⁴⁻³⁸ were full manuscripts, and two studies^{33,39} were only available as abstracts. No studies with comparison or control groups were identified. Study publication dates ranged from 2020 to 2022.

Chatbot: ItRunsInMyFamily

ItRunsInMyFamily chatbot (version 3) was described in two studies.^{34,35} This web-based chatbot was developed at the Medical University of South Carolina and the University of

Utah and was designed to collect family history and assess risk for hereditary cancer. The chatbot is capable of being accessed via browser and is designed for use on mobile phones without need for software download. A personalized risk assessment is provided to users completing the tool. Both studies report on data from use of the chatbot in a nonclinical setting, in which the chatbot was promoted using online marketing campaigns on social media.

Chatbot: Genetics Information Assistant

The Genetics Information Assistant (GIA) was described in two studies.^{33,38} GIA is a web-based tool accessed via computer, tablet, or smartphone and was developed by Clear Genetics Inc (San Francisco, CA) and later acquired by Invitae Corporation.⁴⁰ Available in English and Spanish, GIA provided both personal and family history-based cancer risk assessment for multiple hereditary cancer syndromes and genetic testing education. Both studies took place in the outpatient clinical setting. In the study by Nazareth et al,³⁸ data collected by GIA could be transferred into patients' respective electronic medical records.

Chatbot: Comprehensive Risk Assessment and Education Program

The Comprehensive Risk Assessment and Education (CARE) program was described in one study,³⁹ characterized as a

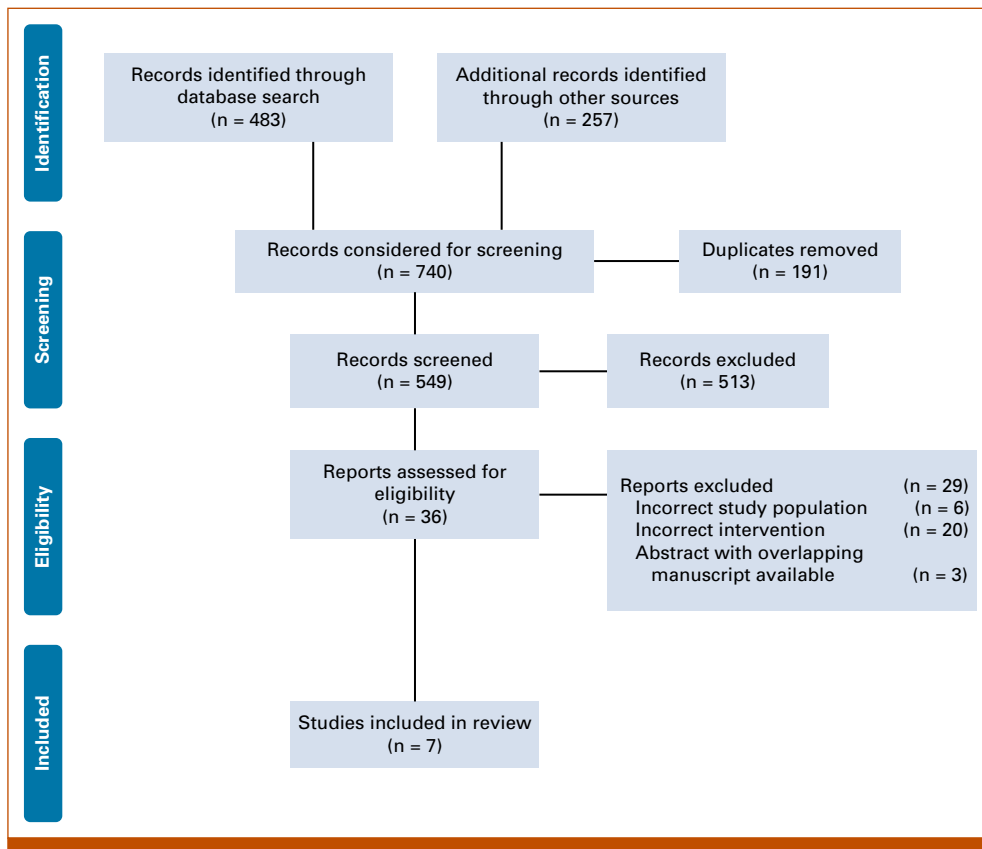


FIG 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram.

TABLE 1. Chatbot Characteristics

Study	Article Type	Described Function	Chatbot	Screening Criteria	Hereditary Cancers Assessed	Genetic Test Panel Used	Recruitment Setting	Invitation	Comparison	Available Languages
Dohany et al ³³	Abstract only	Risk assessment	Chatbot by Clear Genetics Inc (San Francisco, CA)	NCCN guidelines	Included <i>BRCA</i> -related breast and/or ovarian cancer, Lynch, and polyposis syndromes	Not specified	28 OB/GYN practices across the United States	Via text message or email before appointment	None	Not specified
Heald et al ³⁷	Manuscript	Risk assessment, counseling	Not specified	Colon cancer risk assessment tool	Hereditary colorectal cancer syndrome	55-gene pan-cancer panel	Colonoscopy appointments at Cleveland Clinic	Via electronic medical record patient portal or letter before colonoscopy	None	English only
Monsour et al ⁶¹	Abstract only	Risk assessment, counseling	CARE program	NCCN guidelines	Hereditary breast, ovarian and pancreatic cancer, Lynch syndrome, and familial adenomatous polyposis	91-gene panel	Rural GI clinic	Via text or email	None	Not specified
Morgan et al ³⁶	Manuscript	Counseling	BRCA Founder OutReach Study chatbot by LifeLink Systems	NA	<i>BRCA</i> pathogenic variant-associated cancers	<i>BRCA1/2</i> Ashkenazi Jewish founder pathogenic variants	Nonclinical	Via study website	None	English only
Nazareth et al ³⁸	Manuscript	Risk assessment, counseling	GIA	NCCN guidelines	Hereditary breast and ovarian cancer, Lynch, and polyposis syndromes	Multigene panels of at least 30 cancer predisposition genes	180 US clinics including OB/GYN, primary care, gastroenterology, oncology, and surgical services	Via link sent to mobile phone or email before appointment	None	English, Spanish
Ritchie et al ³⁴	Manuscript	Risk assessment	ItRunsInMyFamily	NCCN guidelines, ACMG guidelines	Hereditary breast, ovarian, pancreatic, Li-Fraumeni, colorectal, and endometrial syndromes	NA	Nonclinical	See Welch et al ³⁵	None	Not specified
Welch et al ³⁵	Manuscript	Risk assessment	ItRunsInMyFamily	See Ritchie et al ³⁴	See Ritchie et al ³⁴	NA	Nonclinical	Via online marketing campaigns through social networking, online advertisements, individual solicitations, and promotional banners	None	Not specified

Abbreviations: ACMG, American College of Medical Genetics; CARE, Comprehensive Risk Assessment and Education; GIA, Genetic Information Assistant; NA, not applicable; NCCN, National Comprehensive Cancer Network; OB/GYN, obstetrics and gynecology.

virtual assistant-based chatbot. CARE was described as performing genetic cancer risk assessment and genetic education for multiple hereditary cancer syndromes, as well as breast cancer risk assessment via the Tyrer-Cuzick model. CARE was used in an outpatient setting, and invitations to complete the chatbot were sent via email or text.

Chatbot: BRCA Founder OutReach Study Chatbot by LifeLink Systems

A chatbot-based digital health tool developed by LifeLink Systems was described in one study³⁶ as being used for study eligibility assessment, consent, and pretest education for specific *BRCA1* and *BRCA2* Ashkenazi Jewish founder pathogenic variants. The chatbot was used in a nonclinical setting, with study recruitment occurring through community outreach.

Chatbot: Unspecified

The fifth chatbot identified in this systematic review was an unspecified chatbot described in one study,³⁷ used to administer the Colon Cancer Risk Assessment Tool and provide pretest genetic counseling for a multigene cancer genetic panel in an outpatient clinical setting. The chatbot was distributed to patients via electronic medical record portal message or mailed letter and accessed via a link.

Risk Assessment

Six studies^{33-35,37-39} evaluated four chatbots with a risk assessment component. Three of the chatbots performed a risk assessment for multiple types of hereditary cancer syndromes: GIA,^{33,38} ItRunsInMyFamily,^{34,35} and CARE.³⁹ The fourth chatbot³⁷ performed risk assessment specifically for hereditary colorectal cancer syndromes. These six studies

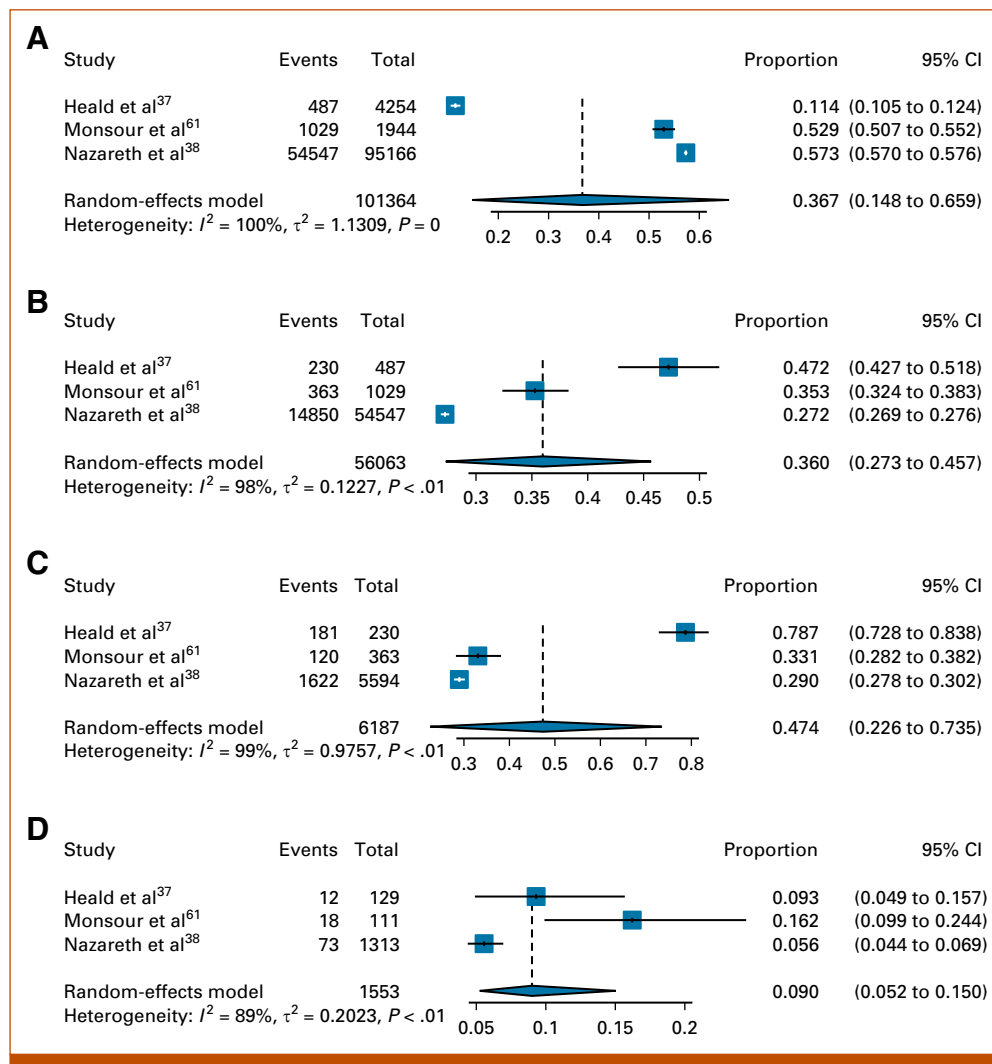


FIG 2. Pooled proportion for (A) completion of chatbot genetic cancer risk assessment, (B) eligibility for genetic testing, (C) having genetic testing ordered, and (D) identifying a cancer-predisposing pathogenic variant.

did not aim to target high-risk patients for inclusion, and instead, the chatbot was distributed to all patients.

Four studies were conducted in outpatient clinics setting, of which 2 studies^{37,39} took place in a GI/colonoscopy clinic setting, one study³³ in obstetrics and gynecology (OB/GYN) practices, and one study³⁸ including a mix of clinics including OB/GYN, primary care, gastroenterology, oncology, and surgical services. No required clinical contact during the risk assessment process was noted, except in the study by Heald et al³⁷ in which any response of unsure to a risk assessment question was followed up with contact by a genetic counseling assistant. Most commonly, the cancer risk assessment was communicated to users via chatbot, except for in the study by Nazareth et al³⁸ in which the cancer risk assessment result was communicated via the clinician.

In three studies³⁷⁻³⁹ with data available for meta-analysis, among individuals approached to use the chatbot, the pooled estimated completion rate for the genetic cancer risk assessment was 36.7% (95% CI, 14.8 to 65.9; [Figs 2A and 3A](#)). Among those completing the chatbot, the pooled estimated rate of eligibility for genetic testing was 36.0% (95% CI, 27.3 to 45.7; [Figs 2B and 3B](#)). Among those eligible, the pooled estimated rate of being ordered for genetic testing was 47.4% (95% CI, 22.6 to 73.5; [Figs 2C and 3C](#)), and among those who received genetic testing with results available, the pooled estimated rate of identification of a cancer-predisposing pathogenic variant was 9.0% (95% CI, 5.2 to 15.0; [Figs 2D and 3D](#)).

Counseling

Four studies³⁶⁻³⁹ described chatbots with the purpose of providing pretest genetic counseling for hereditary cancer syndromes. Three studies³⁷⁻³⁹ were conducted in an outpatient clinical setting with all-comer patients, without intentional enrichment for high-risk patients. In the study by Nazareth et al,³⁸ the GIA chatbot provided genetic testing education, though risk disclosure and discussion of next steps was performed by the clinician. In the study by Heald et al,³⁷ the chatbot provided pretest education and counseling for hereditary cancer syndrome genetic testing and offered the option to proceed with genetic testing. No mandatory counseling by a health care professional was required to proceed with genetic testing, although all requests were reviewed by genetic counselors. In the study by Monsour et al,⁶¹ the CARE chatbot was described as providing pretest education, without specification of the involvement of a clinician.

One study³⁶ was conducted in a nonclinical setting with recruitment through community outreach, in which patients with a known risk factor (≥ 1 grandparent of Ashkenazi Jewish ancestry) were invited to participate. In this study, the chatbot provided pretest education and consent, which was described as approximating pretest counseling. Genetic testing for *BRCA1* and *BRCA2* Ashkenazi Jewish founder

pathogenic variants was offered, and no mandatory counseling by a health care professional was required to proceed with genetic testing.

Two studies^{37,38} reported the proportion of individuals completing the education or counseling portion of the chatbot. Heald et al³⁷ reported that all 215 patients who screened eligible for genetic testing completed the education section. Nazareth et al³⁸ reported that 71.4% of the 61,070 users who engaged with the chatbot completed the education section.

Regarding effectiveness of pretest education and counseling by a chatbot, Morgan et al³⁶ administered an investigator-designed *BRCA1/BRCA2* genetic testing knowledge survey and reported that genetic knowledge was high after pretest education with an average of 90% of items answered correctly. No other studies evaluated knowledge uptake or understanding after pretest genetic counseling.

Study Population Characteristics

Two studies^{37,38} provided comprehensive data regarding patient age, race and ethnicity, and sex. In the study by Heald et al,³⁷ of the 506 patients who opened the chatbot, 86% were White, and 96% were of non-Hispanic ethnicity. The authors found that individuals who were female, who were of White race, or who were delivered the chatbot link via the portal were more likely to open the chatbot and that individuals of Black race were less likely to open the chatbot. The authors found no difference in age among those opening the chatbot versus those not opening the chatbot. In the study by Nazareth et al,³⁸ 58.8% of the 61,070 chatbot users were described as non-Hispanic White. The authors noted that individuals who used the chatbot were significantly younger than nonusers.

Patient Satisfaction

Two studies^{33,38} included participant ratings, both of which used the GIA chatbot. In the study by Dohany et al,³³ respondents rated their experience using GIA as an average of 4.6 of 5. In the study by Nazareth et al,³⁸ participants were asked to rate their satisfaction with GIA, and the average satisfaction rating was 4.6 of 5.

In the study by Morgan et al,³⁶ participants using the chatbot by LifeLink Systems were asked about their interest in using a chatbot again in a research study or health care encounter, and participants indicated an average interest of 7.2 of 10.

Time

One study³⁸ evaluated average chatbot time burden. Nazareth et al³⁸ noted a mean chatbot use duration of 15.4 minutes when activity gaps were excluded with a standard deviation of 2 hours and 56 minutes.

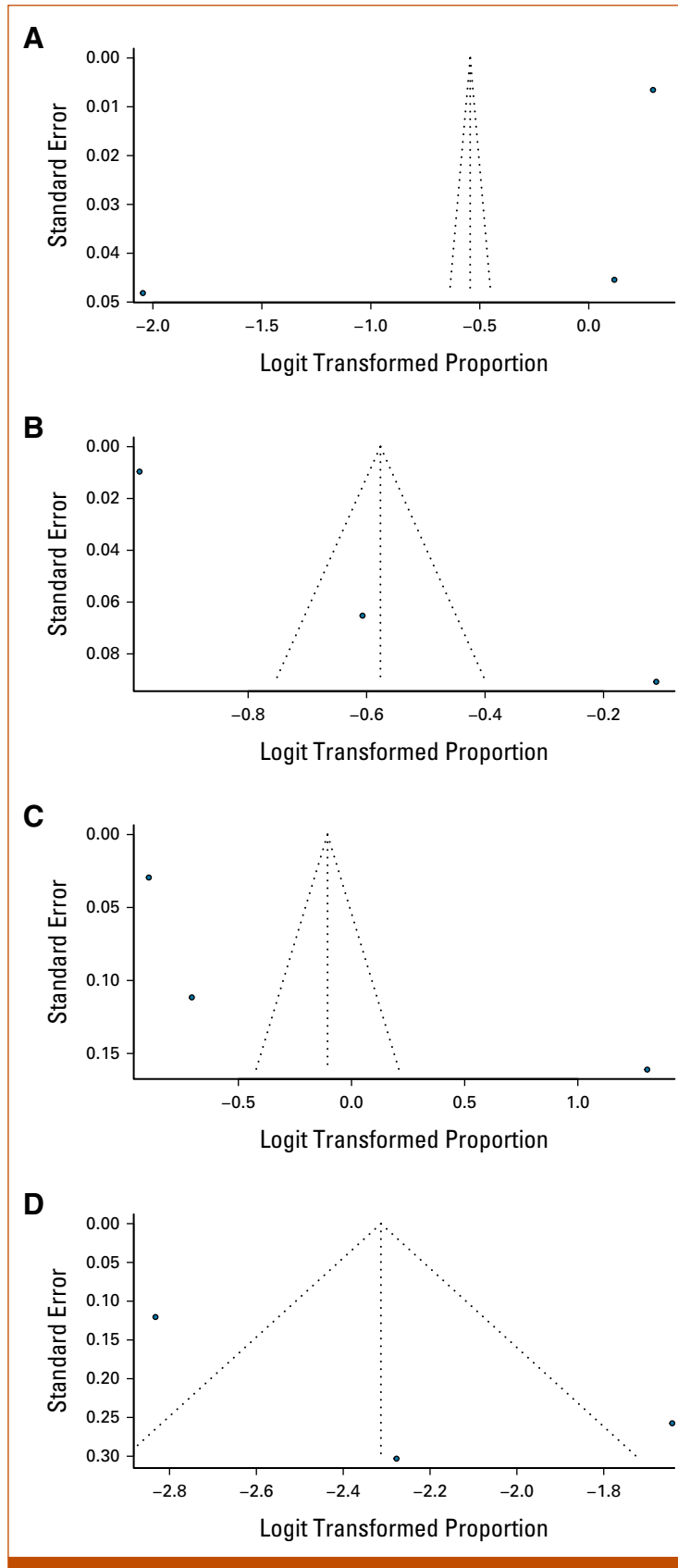


FIG 3. Funnel plots for (A) completion of chatbot genetic cancer risk assessment, (B) eligibility for genetic testing, (C) having genetic testing ordered, and (D) identifying a cancer-predisposing pathogenic variant.

Language Availability

Three³⁶⁻³⁸ studies specified the language(s) in which the chatbot was available or language criteria for study inclusion. In the study by Nazareth et al,³⁸ the GIA chatbot was available in English and Spanish while two studies^{36,37} were limited to English speaking or reading participants.

Risk of Bias

All included studies were evaluated for risk of bias in design, conduct, and analysis relevant to outcomes of interest using the Joanna Briggs Institute cross-sectional study critical appraisal tool (Data Supplement, Material S3). All studies were deemed eligible for inclusion in this systematic review. The most common source of potential bias was a lack of detailed description of study participants or setting and a lack of discussion of confounding factors, particularly among studies for which only abstracts were available.

DISCUSSION

Seven studies featuring five chatbots which performed genetic cancer risk assessment or counseling were identified in this systematic review and meta-analysis. The earliest identified studies were published in 2020, suggesting that the use of chatbots for genetic cancer risk assessment and pretest counseling for cancer genetic testing remains in its infancy. Chatbots were most frequently used for genetic cancer risk assessment with or without additional pretest cancer genetic testing counseling and education. Several chatbots were designed to streamline the screening and testing process independently of a clinician, though others required the intervention or involvement of a health care professional. The pooled estimated completion rate for the genetic cancer risk assessment portion of the chatbot was 36.7%. Although data were limited, evaluation of patient satisfaction with the chatbot appeared to be favorable. Inspection of funnel plots for all pooled proportions indicate a uniform distribution of smaller and larger studies, as well as equal representation of studies with positive and negative results relative to summary estimates. Although this suggests that there is no publication bias among the studies included in our meta-analysis, this assessment should be interpreted with caution given the overall small number of studies quantitatively synthesized.

Heterogeneous patient populations, study settings, and chatbot function as well as the lack of comparison groups and varying degrees of clinician involvement preclude generalizations about the function and effectiveness of evaluated chatbots for risk assessment and pretest counseling. Multiple barriers throughout the process of genetic cancer risk assessment and genetic testing hinder the widespread utilization of associated genetic services. Providers may fail to recognize individuals with elevated risk for cancer-predisposing mutations, either because of lack of pertinent history collection or failure to recognize relevant

criteria, particularly as family history assessment is associated with a large time burden.^{20,41} Additionally, once identified as high risk, patients may fail to receive referrals to genetic specialists or fail to follow through with referrals once made.^{42,43} Through collection of personal and family history and innate analysis capability, chatbots may serve to alleviate clinicians of the time burden and complexity of genetic cancer risk assessment. Furthermore, the interactive nature of chatbots, combined with their ability to deliver tailored educational content, may enhance patient engagement and understanding of genetic testing, as well as mitigate language barriers.

The studies identified in this review inconsistently reported patient demographic and characteristics and thus comprehensive data on user characteristics are lacking. In the two studies that reported race and ethnicity, the majority identified as non-Hispanic White. Previous literature has shown that racial and ethnic minorities, individuals with public insurances, and non-English speaking individuals are underrepresented in the field of genetics and are less likely to receive genetic services.^{8,41,44-48} Differences in technology utilization by age, race, income, geographic location, and literacy levels have also previously been demonstrated.⁴⁹⁻⁵³ In this systematic review, few studies reported on characteristics between chatbot users and chatbot nonusers, precluding an understanding of which populations may be marginalized in settings where chatbot genetic cancer risk assessment could serve as standard of care.

While chatbots offer the possibility to increase access to genetic services by providing a standardized assessment, it remains unclear at this time whether the application of artificial intelligence within medicine exacerbates or reduces disparities.⁵⁴⁻⁵⁷ The potential for chatbots to be available in the preferred language(s) of the patient population or designed for specific health literacy levels suggest that such technology may mitigate language barriers which have been shown to contribute to health disparities, particularly with regard to genetic services.^{48,58} Collection of patient demographic data, including preferred language, insurance status, race, and ethnicity, will serve to elucidate the impact of chatbot use for genetic cancer risk assessment and counseling on existing disparities. An understanding of populations at risk for chatbot incompleteness may also allow for modification of technology to increase tool accessibility for those populations.

Several trials, including randomized and controlled trials, related to the use of chatbot for genetic cancer risk assessment and pretest counseling for cancer genetic testing are listed as active or recruiting on ClinicalTrials.gov: Chatbot to Maximize Hereditary Cancer Genetic Risk Assessment (ClinicalTrials.gov identifier: [NCT05562778](#)), OPTimizing Treatment Focused Genetic Testing IN Cancer (OPT-IN; ClinicalTrials.gov identifier: [NCT04066361](#)), and Effectiveness of AI Genetic Counseling Program versus In-person Genetic Counseling in Breast Cancer (ClinicalTrials.gov identifier: [NCT04354675](#)).

Similarly, another ongoing study, Effectiveness of AI Genetic Counseling Program versus In-person Genetic Counseling in Breast Cancer (ClinicalTrials.gov identifier: [NCT04354675](#)), is designed to compare a chatbot with a genetic counselor for pretest genetic counseling. As artificial intelligence and natural language processing technology continue to advance, the integration of chatbots into genetic cancer risk assessment and pretest genetic counseling holds promise for improving patient care and outcomes, and the evaluation of chatbots in comparative studies will provide valuable data.

The use of chatbots for genetic cancer risk assessment and counseling raises ethical and legal concerns. Ensuring data privacy and confidentiality is crucial, as well as obtaining

informed consent for data usage and communication.⁵⁹ Stringent data encryption and storage protocols can safeguard privacy, in addition to a transparent consent process for patients using chatbots. Liability issues also need addressing, with clear delineation of responsibilities between chatbots and health care providers.⁶⁰ Regular algorithm audits and human oversight could ensure accuracy, alongside incorporation of disclaimers regarding the chatbot's limitations to minimize liability concerns. Collaborative efforts between medical professionals, regulatory bodies, and technology developers are essential to create robust legal frameworks that mitigate these challenges and ensure safe and effective integration of chatbots in clinical practice for hereditary cancer risk assessment.

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EQUAL CONTRIBUTION

R.N.S. and M.K.F. contributed equally to this work as senior authors.

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I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/cci/author-center.

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